

Sarvesh Prasanth Venkatesan

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EDUCATION

University of Southern California	Los Angeles, CA
<i>M.S. in Electrical and Computer Engineering - Machine Learning & Data Science</i>	2025-2027
Vellore Institute of Technology	Vellore, India
<i>B.Tech. in Electronics and Communication Engineering</i>	2019 - 2023

EXPERIENCE

Stochastic Robotics Lab Indian Institute of Science	Bengaluru, India
<i>Project Associate Advisor: Prof. Shishir Kolathaya</i>	July 2024 – July 2025

- Incorporated various locomotion skills using PPO and curriculum-learning for Stoch5 robot in Isaac Gym. [\[Video\]](#)
- Added rewards based on relaxed log-barrier function to enforce gait parameters such as step frequency, joint position, and body height.
- Created a low-level hardware interface for efficient position control to infer policies on the robot.
- Integrated NVIDIA Warp into Isaac Gym, reducing depth and LiDAR frame computation time by 5×.
- Collaborated on a supervised learning framework for generalist quadruped locomotion using offline proprioceptive data. [\[Link\]](#)

Mobile Robotics Lab Indian Institute of Science	Bengaluru, India
<i>Project Associate Advisors: Prof. Debashish Ghose & Prof. Shishir Kolathaya</i>	Sep 2023 – June 2024

- Worked on a collaborative multi-agent system for agriculture. [\[Video\]](#)
- Developed LocalizeSORT, a multi-object tracking framework with world-coordinate association, achieving 15% higher HOTA and 30% higher yield prediction accuracy. [\[Link\]](#)
- Implemented a vision-based navigation stack for both the aerial and quadruped robot.
- Built a custom 2-R manipulator and integrated it with quadruped robot for spraying capabilities.
- Deployed a centralized multi-agent pipeline with a client-server framework, maintaining functionality even after communication loss.

Centre for Non-Destructive Evaluation Indian Institute of Technology Madras	Chennai, India
<i>Intern Advisors: Prof. Prabhu Rajagopal</i>	Sep 2022 – July 2023

- Built a Shopping Assistance Robot with a mobile base and manipulator. [\[Video\]](#)
- Implemented RRT-based motion planning for a 6-DOF manipulator for automating grocery pickup tasks.
- Deployed real-time SLAM with obstacle avoidance using Lidar and depth camera in cluttered store environment.
- Contributed to the development of 3-D object tracking framework for store inventory management.

PREPRINTS

- Srihari Vemuru, Kumar Ankit, **Sarvesh Prasanth Venkatesan**, Debashish Ghose, Shishir Kolathaya. Yield Prediction and Counting Using World Coordinates. Under Review at *Computers and Electronics in Agriculture*. [\[Link\]](#)
- Narayanan PP, **Sarvesh Prasanth Venkatesan**, Srinivas Kantha Reddy, Shishir Kolathaya. GRoQ-LoCO: Generalist and Robot-agnostic Quadruped Locomotion Control using Offline Datasets. *arXiv preprint arXiv:2505.10973* (2025). [\[Link\]](#)

SKILLS

Programming:	Python, C++, C, R, Verilog.
Robotics:	ROS, MoveIt, Isaac Gym, MuJoCo, PX4, ArduPilot, PyTorch, OpenCV.
Software:	Docker, Git, Conda, Fusion360, MATLAB, LaTeX, Ubuntu Linux.
Hardware:	Unitree Go1, UR5, Husky, PiPER, Nvidia Jetson, Intel RealSense, Moteus.